

Lab: Stats Unit

Import necessary libraries

```
In [1]: import pandas as pd
from scipy.stats import ttest_ind, chisquare, mannwhitneyu, f_oneway
import statsmodels.api as sm
from statsmodels.formula.api import ols
import matplotlib.pyplot as plt
```

Step 1: Read mortality.csv

```
In [2]: mortality_data = pd.read_csv("mortality.csv")
```

```
In [3]: mortality_data.head()
```

```
Out[3]:
```

	Year	Age Group	Death Rate
0	1900	1-4 Years	1983.8
1	1901	1-4 Years	1695.0
2	1902	1-4 Years	1655.7
3	1903	1-4 Years	1542.1
4	1904	1-4 Years	1591.5

```
In [4]: print(mortality_data.columns)
```

```
Index(['Year', 'Age Group', 'Death Rate'], dtype='object')
```

Step 2: Select the mortality rates for the '5-9 Years' and '10-14 Years' age groups and assign these each to a new variable.

```
In [5]: # Select the mortality rates for the '5-9 Years' and '10-14 Years' age groups
mortality_5_9 = mortality_data[mortality_data['Age Group'] == '5-9 Years']['Death Rate']
mortality_10_14 = mortality_data[mortality_data['Age Group'] == '10-14 Years']['Death Rate']
```

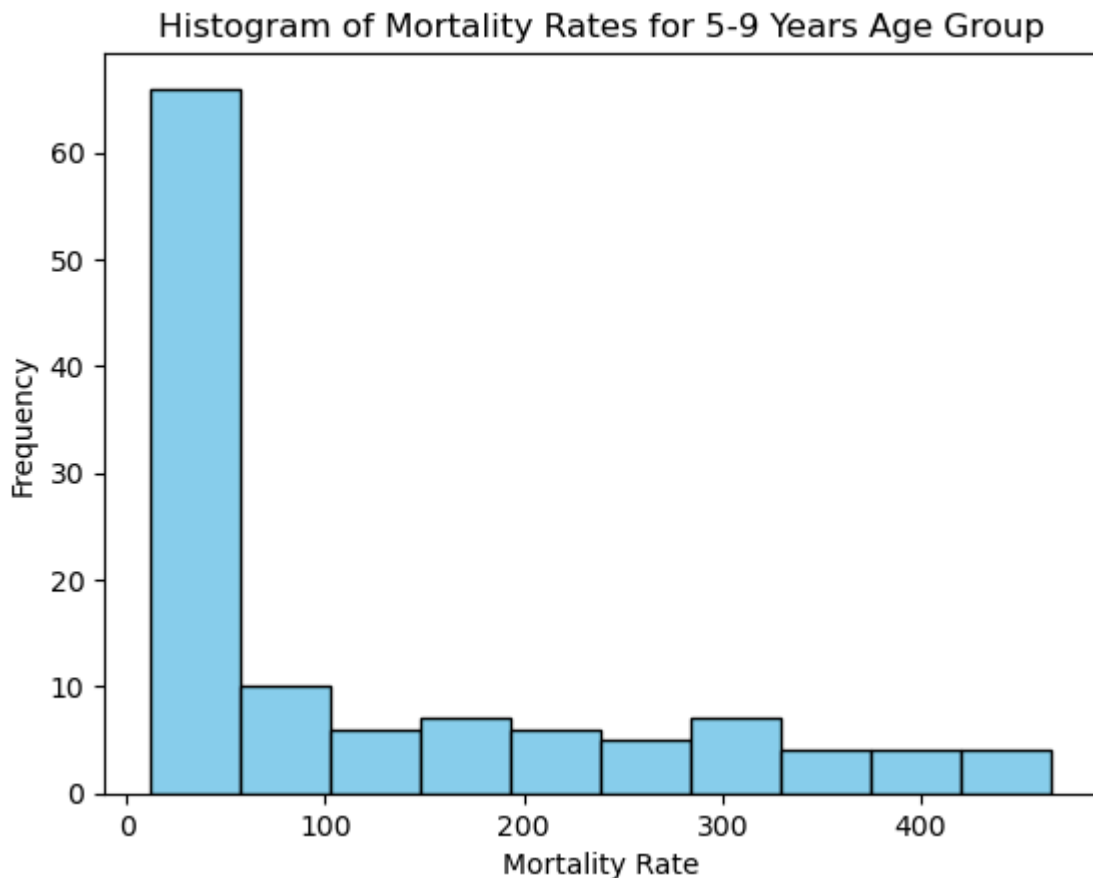
Step 3: Run a t-test to see if these are significantly different

```
In [6]: # Run a t-test
t_stat, p_value = ttest_ind(mortality_5_9, mortality_10_14)
if p_value < 0.05:
    print("Reject the null hypothesis. There is a significant difference between the two groups.")
else:
    print("Accept the null hypothesis. There is no significant difference between the two groups.")
```

Accept the null hypothesis. There is no significant difference between the mortality rates of 5-9 years and 10-14 years age groups.

Step 4: Make a histogram of the 5-9 year old age group to see if it looks normally distributed (bell-curve shaped)

```
In [7]: # Make a histogram of the 5-9 year old age group
plt.hist(mortality_5_9, bins=10, color='skyblue', edgecolor='black')
plt.title('Histogram of Mortality Rates for 5-9 Years Age Group')
plt.xlabel('Mortality Rate')
plt.ylabel('Frequency')
plt.show()
```



Step 5: Run a non-parametric Mann Whitney U Test to see if you get a different result than the t-test

```
In [8]: # Run a non-parametric Mann Whitney U Test
u_stat, p_value_mannwhitneyu = mannwhitneyu(mortality_5_9, mortality_10_14)
if p_value_mannwhitneyu < 0.05:
    print("Reject the null hypothesis. There is a significant difference between")
else:
    print("Accept the null hypothesis. There is no significant difference between")
```

Accept the null hypothesis. There is no significant difference between the mortality rates of 5-9 years and 10-14 years age groups.

Step 6: Select the mortality rates for the 15-19 year old age group and assign it to a new variable

```
In [9]: # Select the mortality rates for the 15-19 year old age group
mortality_15_19 = mortality_data[mortality_data['Age Group'] == '15-19 Years'][''
```

Step 7: Use all three groups and run an ANOVA to check if there are any differences between these groups

```
In [10]: f_stat, p_value_anova = f_oneway(mortality_5_9, mortality_10_14, mortality_15_19)
# Write a comment indicating whether there are any differences between these groups
if p_value_anova < 0.05:
    print("Reject the null hypothesis. There are significant differences between the groups")
else:
    print("Accept the null hypothesis. There are no significant differences between the groups")
```

Reject the null hypothesis. There are significant differences between the mortality rates of the age groups.

Step 8: We rolled a six-sided die 60 times, and got some weird results. Use this data to run a chi-squared test to see if we reject the null hypothesis that the frequency counts for each face of the die conform to the expected uniform distribution.

```
In [11]: # Data for chi-squared test
observed = [8, 8, 8, 8, 8, 20]
expected = [10, 10, 10, 10, 10, 10]
```

```
In [12]: # Run a chi-squared test
chi_stat, p_value_chi = chisquare(observed, expected)
# Write a comment indicating whether we reject the null hypothesis
if p_value_chi < 0.05:
    print("Reject the null hypothesis. The frequency counts for each face of the die do not conform to the expected uniform distribution.")
else:
    print("Accept the null hypothesis. The frequency counts for each face of the die conform to the expected uniform distribution.")
```

Reject the null hypothesis. The frequency counts for each face of the die do not conform to the expected uniform distribution.

1. Select the Years for the 15-19 Years age group and assign it to a variable named 'x'

```
In [13]: # Select the Years for the 15-19 Years age group
x = mortality_data['Year']
```

2. Select the DeathRate for the 15-19 Years age group and assign it to a variable named 'y'

```
In [14]: # Select the DeathRate for the 15-19 Years age group
y = mortality_data['Death Rate']
```

3. Use the OLS model from Chapter 10's lecture notebook to test whether there is a linear relationship between Year and DeathRate for the 15-19 year old age group in the mortality data.

```
In [15]: # Add a constant term to the predictor variable x
```

```
x = sm.add_constant(x)
```

```
In [16]: # Use the OLS model to test for linear relationship
model = sm.OLS(y, x).fit()
print(model.summary())
```

```

                    OLS Regression Results
=====
Dep. Variable:      Death Rate      R-squared:                0.387
Model:              OLS             Adj. R-squared:           0.385
Method:             Least Squares   F-statistic:             298.8
Date:              Wed, 24 Apr 2024  Prob (F-statistic):       2.86e-52
Time:              22:32:02         Log-Likelihood:          -3262.7
No. Observations:   476            AIC:                     6529.
Df Residuals:       474            BIC:                     6538.
Df Model:           1
Covariance Type:    nonrobust
=====
                    coef    std err          t      P>|t|      [0.025    0.975]
-----
const      1.058e+04    600.994     17.604     0.000    9399.055    1.18e+04
Year       -5.3022      0.307    -17.286     0.000     -5.905    -4.699
=====
Omnibus:            350.796   Durbin-Watson:           0.065
Prob(Omnibus):      0.000   Jarque-Bera (JB):        4064.313
Skew:               3.220   Prob(JB):                 0.00
Kurtosis:           15.785   Cond. No.                 1.12e+05
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.12e+05. This might indicate that there are strong multicollinearity or other numerical problems.